Hands-on Session 1: Sub-Halo Abundance Matching

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ABSTRACT

In this report, we investigated the relationship between galaxies and their host dark matter halos in the local Universe, which is dominated by dark matter. Utilizing data from the Millennium Simulation (Springel et al. (2005)) and observational stellar mass functions, we applied the Sub-Halo Abundance Matching (SHAM) technique to derive the stellar to halo mass relation at redshift $z = 0$. Our analysis reveals a fundamentally non linear connection between the two masses, characterized by a star formation efficiency that peaks at a halo mass of $M_h \approx 10^{11.7} M_\odot$. We showed how this efficiency remains consistently below the universal baryon fraction, highlighting the impact of physical feedback mechanisms that oppose to star and galaxy formation. Finally, we compared our results with an hydrodynamic simulation, in order to demonstrate that the SHAM method accurately captures the general trend of galaxy formation physics. Lastly, we discussed the observed decline in efficiency at both low and high mass scales in our analysis taking into account physical processes such as stellar feedback and shock heating, confirming the essential role of baryonic physics in regulating galaxy evolution.

1 INTRODUCTION

The Universe we live in is dominated by dark matter, which accounts for approximately 85% of the total matter content of the Universe. Although dark matter has not been directly detected yet, its existence is strongly supported by a wide range of astrophysical and cosmological observations, such as its gravitational effects on visible matter and background radiation. Dark matter plays a fundamental role in the cosmic structure: it dictates the dynamics of the Universe and is essential in the formation and evolution of galaxies.

The origin of these structures can be traced back to the primordial Universe. Following recombination (at $z \approx 1100$), when radiation decoupled from matter, the Universe was characterized by density fluctuations. these perturbations initially expanded with the Hubble flow but eventually detached, entered a non-linear regime and caused dark matter to collapse under its own gravity to form massive and approximately spherical structures known as halos.

These dark matter halos represent the potential wells where ordinary baryonic matter (gas) can later accumulate and form galaxies: as gas falls into the halo, it eventually cools down to form stars and galaxies at the center of the potential well. Consequently, the physical properties of galaxies, such as their stellar mass (M_*), are expected to be strongly correlated with the mass of their hosting halo (M_h). In fact, every galaxy we observe is embedded within an extended dark matter halo that is significantly larger than the visible component of the galaxy itself. For instance, a galaxy like the Milky Way extends for $3 \cdot 10^4$ pc while its surrounding halo reaches sizes of hundreds of thousands of parsecs. This clear separation between the luminous and dark components highlights the dominant role of dark matter in galactic dynamics.

The standard cold dark matter (CDM) paradigm states that the formation of dark matter halos proceeds in a hierarchical manner: smaller halos form first and later merge to build progressively larger structures (figure 1). This scenario successfully explains many observed features of galaxies and large-scale cosmic structures and provides the framework for understanding the distribution of galaxies we observe today.

To study the connection between galaxies and halos, numerical simulations are an essential tool. Hydrodynamic simulations try to model the complex physics of structure formation by including both baryonic and dark matter processes, but this makes them computationally expensive and limited to small volumes. A powerful and less expensive alternative is to use dark matter only simulations, where galaxies are associated with their halos through analytic methods.

In this hands-on session, we adopted the Sub-Halo Abundance Matching (SHAM) method, which assumes a monotonic relationship between the galaxy's stellar mass M_* and the halo mass M_h. We applied SHAM to the Millennium Simulation (Springel et al. (2005)), a dark matter only simulation, to derive the stellar-to-halo mass relation and the star formation efficiency. Finally, we will compare our statistical results with the hydrodynamical predictions from a simulation known as Illustris TNG (Springel et al. (2018)).

2 METHODS AND DATA ANALYSIS

2.1 Datasets

In this analysis, we used two different datasets to reconstruct the relationship between the galaxy's stellar mass and the dark matter halo mass in the local Universe, at redshift $z = 0$.

As we mentioned in the introduction, for the dark matter component we used a catalog from the Millennium Simulation (Springel et al. (2005)), which is a cosmological simulation that traces the evolution of dark matter in a periodic box of side L=500 Mpc/h_0. For this study, we selected a sub volume of 150^3 Mpc with and Hubble parameter of $h_0 = 0.7$ and extracted the virial mass for each object.

To model the baryonic component, instead, we adopted the observational data from Baldry et al. (2012), which provide the galaxy Stellar Mass Function (SMF), the number density of galaxies as a function of stellar mass, in the local Universe. This Stellar Mass Function was then converted into an expected number of galaxies for our simulation volume by uniformly sampling its bins, generating a catalog of synthetic stellar masses.

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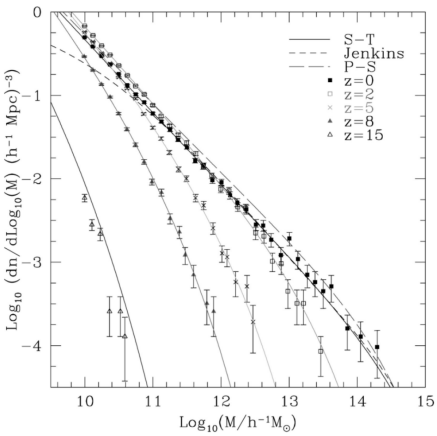

Figure 1. Evolution of the halo mass function from high redshift (z=15) to the present day (z=0), as predicted by CDM models. The plot illustrates the hierarchical growth of structures: massive halos only exist at lower redshifts, formed through the merge of smaller progenitors. Figure from Reed et al. (2003)

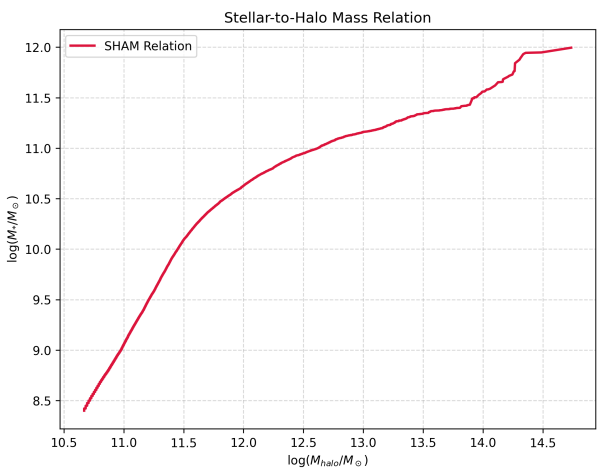

Figure 2. Plot of the relationship between dark matter halo mass (x axis) and galaxy's stellar mass (y axis) in log-log scale, derived with the SHAM technique. The trend of the curve is nearly linear for lower halo masses and changes slope to a flatter one from $M_h \approx 10^{11.7} M_\odot$.

2.2 Sub-Halo Abundance Matching (SHAM)

This technique is based on the assumption that the most massive galaxies live in the most massive halos. Consequently, in order to link M_* and M_h taking this assumption into account, we ranked both the simulated dark matter halos and the synthetic galaxies in descending order and performed a one to one assignment.

The derived stellar to halo mass relation is shown in figure 2. The trend of the function is in fact monotonic (M_* is growing with M_h) but we notice that the relationship is not linear, which implies that the stellar mass does not simply scale as a constant fraction of the halo mass. The slope varies significantly: it is steeper at the low mass halos and becomes flatter for massive halos. This behavior suggests that the efficiency with which baryons are converted into stars depends strongly on the halo mass.

To investigate this further, we computed the baryon conversion efficiency (defined as the ratio M_*/M_h) and plotted it against the halo mass, as shown in figure 3. The efficiency is not uniform but presents

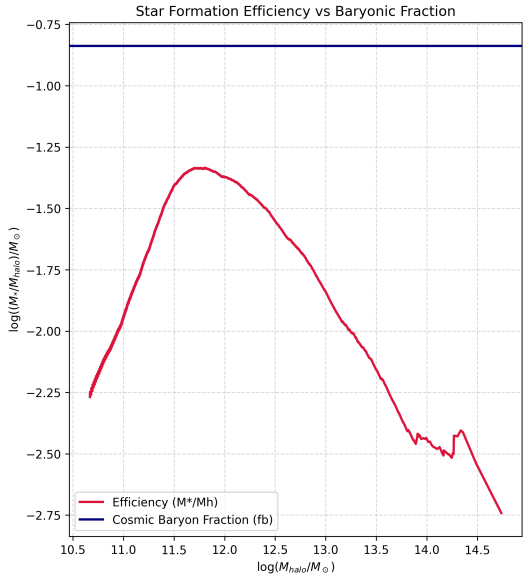

Figure 3. The red line represent the star formation efficiency plotted as a function of the halo mass. The horizontal blue line indicates the cosmic baryon fraction, which sets the theoretical upper limit for the efficiency of baryon conversion into stars. The derived efficiency is generally low, peaking at a halo mass of $M_h \approx 10^{11.7} M_\odot$, and dropping significantly at both higher and lower masses.

a curved shape. At lower masses, the efficiency is low but it's increasing until it reaches a peak at a halo mass of approximately $M_h \approx 10^{11.7} M_\odot$. After this value, as expected from the trend of the M_* vs M_h curve, the efficiency decreases again.

Furthermore, we compared this ratio to the cosmic baryon fraction f_b, given by the ratio between baryonic matter density ($\Omega_f = 0.045$) and the total matter density of the local Universe ($\Omega_M = 0.31$) and shown as the blue line in figure 3. This comparison is useful because f_b represents the maximum amount of baryonic matter available in the Universe relative to the total matter and from our analysis we observe that the efficiency curve lies constantly below the cosmic baryon fraction, suggesting that galaxy formation is an inherently inefficient process: even at the peak of efficiency, only a fraction of the available baryons gets converted into stars and galaxies.

2.3 Cosmological Hydrodynamic Simulations

In the introduction, we briefly mentioned that, apart from statistical dark matter only simulations, there are more sophisticated hydrodynamic simulations. These are considered more complex because they track not only the evolution of dark matter halos but also the formation and evolution of the galaxies inside them. To validate the results obtained through the Sub-Halo Abundance Matching method, we compared the star formation efficiency derived from SHAM with the corresponding quantity obtained from a hydrodynamic simulation.

Since baryonic processes are involved and our understanding of all these phenomena is still incomplete, different hydrodynamic simulations can lead to different predictions. For our comparison, we used the Illustris TNG simulation (Springel et al. (2018)), which provides, among many quantities, the stellar mass and halo mass for individual galaxies and their hosting halos.

We computed the mean stellar to halo mass relation from the TNG catalog and overlapped it on our SHAM results (figure 4). The comparison indicates that the two approaches are consistent in terms of

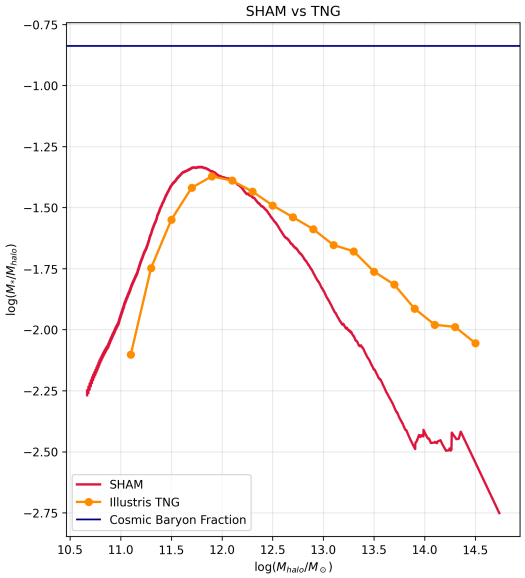

Figure 4. Star formation efficiency as a function of halo mass. The red curve shows the relation derived with SHAM while the orange curve shows the Illustris TNG hydrodynamical simulation. The dark blue horizontal line represents the universal cosmic baryon fraction limit.

the overall shape, confirming that the statistical SHAM technique effectively captures the essential physics of galaxy formation and dark matter halos.

The behavior of the baryon conversion efficiency is roughly similar in both models. Both predict a peak in star formation efficiency at a characteristic halo mass of approximately $M_h \approx 10^{11.7} M_\odot$, although the TNG model places this characteristic mass slightly higher. In addition, in both cases, the peak efficiency remains well below the cosmic baryon fraction, indicating that only a fraction of baryons is converted into stars even at the most efficient halo mass.

At lower halo masses (to the left of the peak), the efficiency is low in both models but rises rapidly with halo mass. After reaching the peak, both models show a decline in efficiency. This is where the largest discrepancy between the two simulations emerges: the SHAM efficiency drops faster than the TNG's. This difference suggests that hydrodynamic simulations predict a higher stellar mass content in massive clusters compared to the purely statistical expectation obtained from sub-halo abundance matching.

Overall, the comparison reinforces the reliability of the SHAM method in reproducing the general trends of galaxy formation efficiency, while also highlighting the small details introduced by baryonic physics that are captured by hydrodynamic simulations.

3 RESULTS

The results obtained from the SHAM analysis and validated by the comparison with the Illustris TNG simulation (Springel et al. (2018)) highlight the non linear nature of the relationship between stellar and halo mass. The most significant result of our analysis is the shape of the star formation efficiency, which increases with the halo mass, peaks at approximately $M_h \approx 10^{11.7} M_\odot$ and declines. This behavior can be physically interpreted as the consequence of interaction between gravitational heating and radiative cooling.

In general, in fact, a galaxy is formed when gas, gravitationally attracted to the potential well of the halo, dissipates its energy and cools down. This cooling accretion process can happen in two ways: "hot mode", when the gas first reaches extremely high temperatures that generate shock waves and consequently cools slowly, and "cold mode", when gas manages to flow directly towards the center of the well without being subjected to shock waves.

Our analysis shows that for massive halos ($M_h > 10^{11.7-12} M_\odot$) the baryon conversion efficiency drops significantly. Theoretically, this regime of masses corresponds to the domain of the "hot mode" accretion. In this regime, gas is heated to the virial temperature and, in order to form galaxies, this gas must cool. However cooling is only effective when the cooling time is shorter than the free-fall time. Given that

$$t_{\rm cool} \propto (1+z)^{-3} \tag{1}$$

$$t_{\rm ff} \propto (1+z)^{-3/2}, \tag{2}$$

we expect cooling to be more effective at high redshift, where smaller galaxies form first (we already mentioned the hierarchical growth of galaxies). Consequently, in more massive halos, more precisely in halos that are more massive than $M \approx 10^{12-13} M_\odot$, the cooling time exceeds the dynamical time, meaning the gas will remain hot and in hydrostatic equilibrium rather than condensing into stars.

This theoretical threshold is very similar to the value we observed in our efficiency plot, for both the SHAM and the hydrodynamic method.

These considerations would also imply that, for low mass halos, cooling is very efficient, since $t_{ff} \gg t_{cool}$, allowing gas to accrete via "cold mode". However in our analysis, for halos that are smaller than $10^{11.7-12} M_\odot$, we observe a steep decline in the efficiency: while the theory suggests that low mass halos should form a large number of stars, our results are indicating that this gas is not efficiently converted into stars.

This low efficiency can be explained with stellar feedback, particularly with Supernovae. In smaller halos, the gravitational potential well is shallow and the energy released by supernovae explosions generates strong galactic winds that can consequently reheat the gas or eject it entirely from the halo, preventing it from forming stars. The fact that our SHAM results are in agreement with the Illustris TNG predictions (in which these feedback processes are taken into account, because this method simulates the behavior of baryonic matter as well), confirms that the feedback mechanisms of suppression by shock waves and supernova winds are regulators of galaxy formation and growth.

4 CONCLUSIONS

In this work, we investigated the connection between dark matter halos and their resident galaxies by applying the Sub-Halo Abundance Matching (SHAM) technique to the Millennium Simulation (Springel et al. (2005)). By linking the dark matter halo masses to the stellar mass function from observational data (Baldry et al. (2012)), we successfully reconstructed the stellar to halo mass relation and the star formation efficiency at redshift $z = 0$.

Our findings confirm that the relationship between M_* and M_h is non linear, indicating that the conversion of baryons into stars is not a constant process but depends strongly on the hosting halo mass. The star formation efficiency is a curve shaped like a bell that peaks at $M_h \approx 10^{11.7} M_\odot$, suggesting that galaxies of intermediate mass are the most efficient at converting their available gas into stars. However, even at this peak, the efficiency remains significantly below the cosmic baryon fraction f_b. The suppression of star formation at

different scales can be attributed to different physical mechanisms: stellar feedback from supernovae winds in low mass halos with shallow potential wells and shock heating in massive halos, where gas accretion shifts to a "hot mode" that slows down efficient cooling.
The general agreement between our SHAM results and the Illustris TNG hydrodynamic simulation (Springel et al. (2018)) validates the dark matter only simulations as a computationally efficient tool. While dark matter provides the "base" for structure formation, the final properties of galaxies are dictated by the complex interplay of baryonic physics. In conclusion, dark matter sets the overall structure of the Universe, but the final properties of galaxies are shaped by baryonic physical processes. The agreement between dark matter only models and hydrodynamic simulations supports the framework of a dark matter dominated Universe and shows that feedback plays a key role in controlling galaxy growth over time.

REFERENCES

Baldry I. K., et al., 2012, Monthly Notices of the Royal Astronomical Society, 421, 621
Reed D. H., O'Grady J. J., Ballou J. D., Frankham R., 2003, in Animal conservation forum. pp 109–114
Springel V., et al., 2005, nature, 435, 629
Springel V., et al., 2018, Monthly Notices of the Royal Astronomical Society, 475, 676